graded_bets

equity strategy | family: equity-baseline | status: validated

What this strategy does

Bet on every stock; the more extreme the surprise, the bigger the bet

Why it should work. Same drift as Extreme Picks but harvested across the whole cross-section for lower variance.

Design notes (from the strategy's source file).

Graded Bets -- every stock gets a bet, bigger the further its surprise is from average.

Trades every decile with weight proportional to SUE-rank distance from the median: broader diversification than the extreme-decile book, at the price of ~4x more positions (which saturates the liquidity and leverage caps and dilutes the high-conviction names).

parameter	value
base_unit_fraction	0.002
max_gross_leverage	1.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	proportional
exposure_cap	open_book
beta_overlay	0.0

Performance

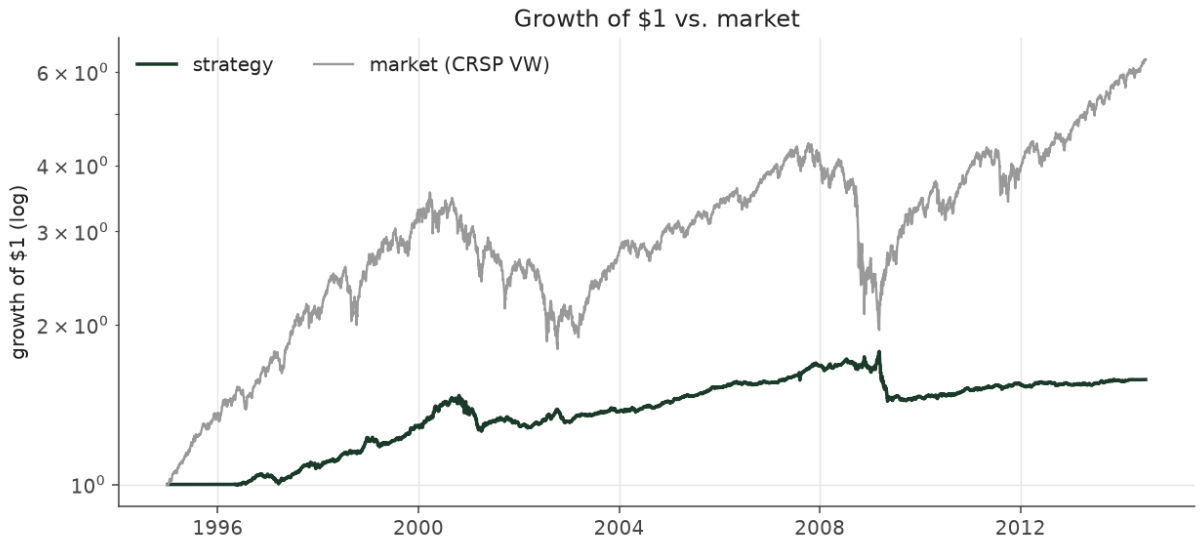
metric	graded bets	market (CRSP VW)
Annualized return	2.37%	9.94%
Annualized volatility	3.91%	19.35%
Sharpe (NAV-pct)	0.62	0.59
Sortino	0.85	
Max drawdown	-19.5%	-55.5%
Calmar	0.12	
Final NAV ($10M start)	$15,766,792	
Worst day	-2.73%	
% positive days	50.1%	
Sharpe 1996-2001	0.96	
Sharpe 2002-2007	1.52	
Sharpe 2008-2013	-0.23	

Correlation of daily returns with the market: **-0.01**.

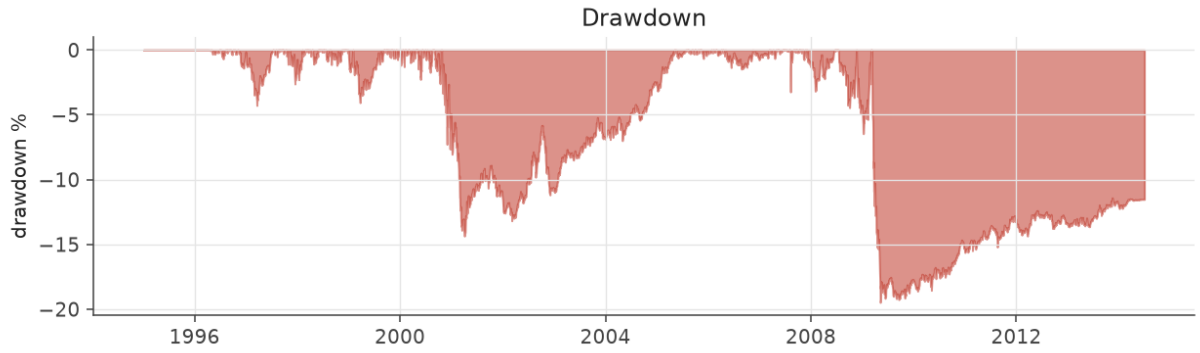
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 0.617

• best return among equity: `market_boost_max` (0.116); this one: 0.0237

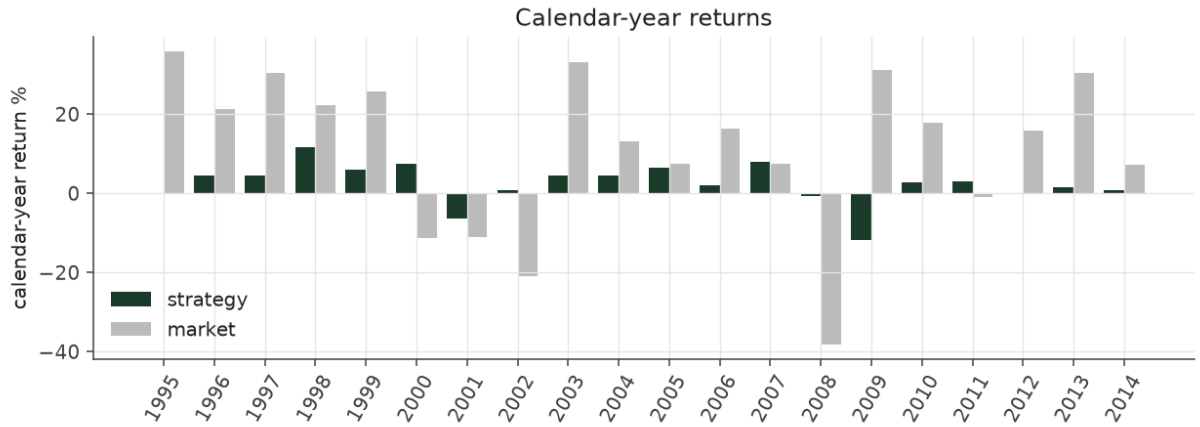
• best Calmar among equity: `calm_rank_tilt` (0.442); this one: 0.121



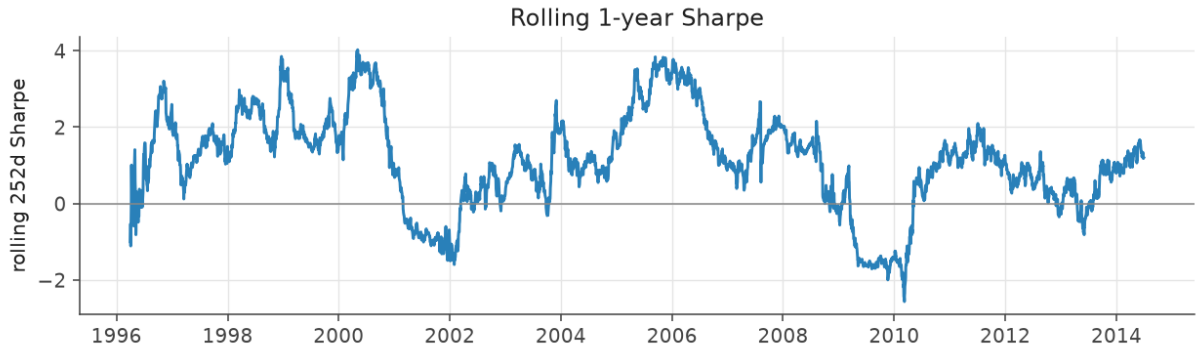
NAV (log scale)



Drawdown from running peak

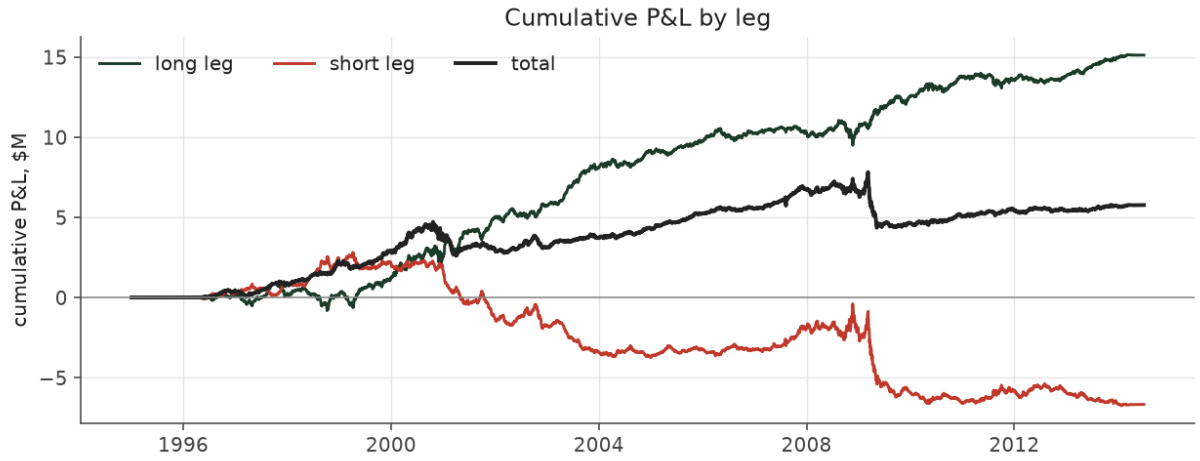


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	15.12	-6.66
P&L after costs ($M)	13.80	-8.04
of which common (average-event) component ($M)	10.32	-10.64
of which excess / PEAD component ($M)	4.80	3.98
excess minus costs ($M)	3.49	2.61
gross return per $ traded (bp)	223.40	-107.45
positions in the money (hit rate)	0.55	0.48
stand-alone Sharpe (leg only)	1.08	-0.42

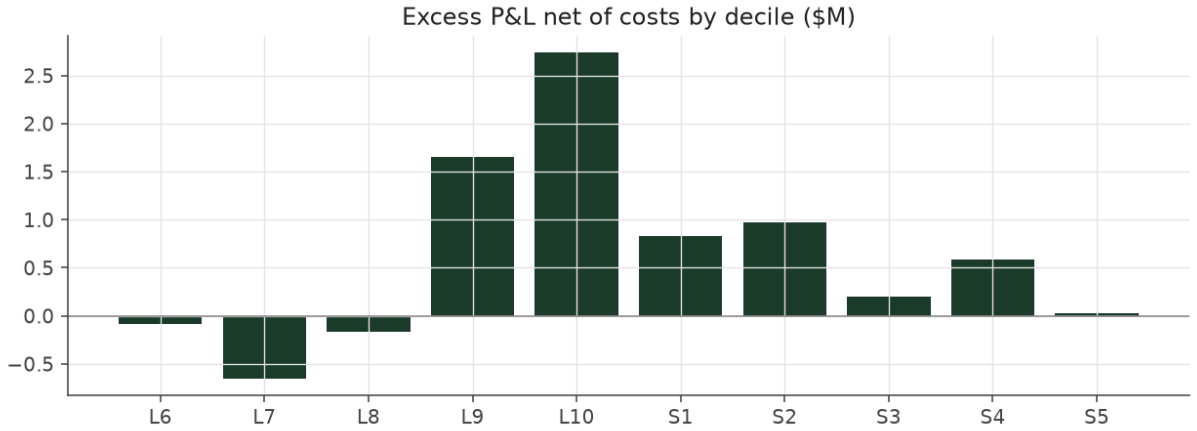
Daily P&L correlation between the two legs: **-0.71**.

• **The short leg loses money in absolute terms** (-6.7M before costs) while the long leg earns +15.1M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 10.3M / 10.6M. Stripping that common component, the surprise signal itself earned +4.8M on the long side and +4.0M on the short side.

• The two legs' daily P&L are -0.71 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.08 long / -0.42 short vs 0.62 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	6	12,092	16.62	0.09	-0.05	0.04	-0.08	-1.4%	-49.68
long	7	20,923	85.25	0.72	-0.49	0.17	-0.66	-10.8%	-77.09
long	8	16,617	125.39	2.03	0.08	0.24	-0.17	-2.8%	-13.44
long	9	18,570	187.17	4.08	2.01	0.36	1.65	27.1%	88.38
long	10	20,136	262.31	6.88	3.25	0.51	2.74	44.9%	104.44
short	1	18,299	230.08	-3.48	1.35	0.53	0.83	13.6%	36.05
short	2	18,907	188.38	-2.23	1.39	0.41	0.97	15.9%	51.48
short	3	15,390	110.41	-1.50	0.44	0.24	0.20	3.2%	17.65
short	4	20,837	79.92	-0.70	0.75	0.17	0.59	9.6%	73.46
short	5	11,143	11.40	-0.12	0.05	0.02	0.03	0.5%	24.41



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,552	127.72	-0.80	0.36	0.52	-0.16	-2.7%	-12.66
2	32,133	252.41	2.07	3.21	0.73	2.48	40.8%	98.39
3	36,789	287.83	2.14	2.72	0.60	2.12	34.8%	73.62
4	40,259	309.95	1.15	1.37	0.48	0.89	14.6%	28.63
5	44,181	319.03	1.21	1.12	0.35	0.77	12.6%	24.02

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	19,144	114.00	0.99	1.90	0.57	1.33	21.9%	117.05
2	32,282	258.60	0.70	2.19	0.78	1.41	23.2%	54.57

3	37,327	292.93	1.98	2.77	0.59	2.19	35.9%	74.63
4	40,479	311.30	1.09	1.31	0.44	0.87	14.3%	28.09
5	43,677	320.05	1.00	0.61	0.32	0.29	4.7%	9.03

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	68,512	438.10	-0.20	2.50	0.88	1.63	26.7%	37.12
41-60d	40,665	317.83	0.14	1.67	0.71	0.96	15.7%	30.15
61-90d	47,915	438.37	3.16	2.12	0.81	1.31	21.5%	29.85
>90d	15,822	102.64	2.68	2.49	0.29	2.20	36.1%	214.49

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	49,863	363.12	2.87	3.52	0.73	2.80	45.9%	77.03
2002-07	58,152	458.18	3.70	4.19	0.92	3.27	53.7%	71.48
2008-13	64,899	475.64	-0.81	1.06	1.04	0.02	0.4%	0.47

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	32,014	245.31	1.09	0.68	0.50	0.18	3.0%	7.37
Chems	3,914	28.15	0.02	0.12	0.05	0.07	1.1%	23.83
Durbl	4,045	30.90	0.17	0.32	0.07	0.26	4.2%	82.71
Enrgy	9,090	63.90	0.56	0.78	0.12	0.65	10.7%	102.25
Hlth	17,523	124.41	0.68	0.70	0.28	0.42	6.9%	33.66
Manuf	16,998	131.12	0.39	0.95	0.27	0.68	11.1%	51.79
Money	25,762	195.29	0.80	1.87	0.40	1.47	24.1%	75.11
NoDur	7,470	55.55	0.15	0.23	0.11	0.12	1.9%	20.98
Other	26,552	196.77	1.12	1.76	0.44	1.31	21.6%	66.85
Shops	17,482	132.16	0.87	1.12	0.27	0.85	14.0%	64.41
Telcm	5,624	43.16	-0.14	0.06	0.08	-0.02	-0.3%	-4.70
Utils	6,440	50.22	0.05	0.20	0.09	0.11	1.8%	21.89

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	56,962	430.97	-0.56	0.47	0.80	-0.33	-5.4%	-7.58
low vol	58,063	431.85	3.41	4.52	0.99	3.53	58.0%	81.82
mid vol	55,419	412.94	2.66	3.53	0.85	2.68	44.0%	64.95

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	2.37%	0.00%
0.5%	0.6	2.15%	0.22%
1.0%	1.3	1.92%	0.45%
2.0%	2.6	1.44%	0.93%
5.0%	6.4	-0.34%	2.71%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.